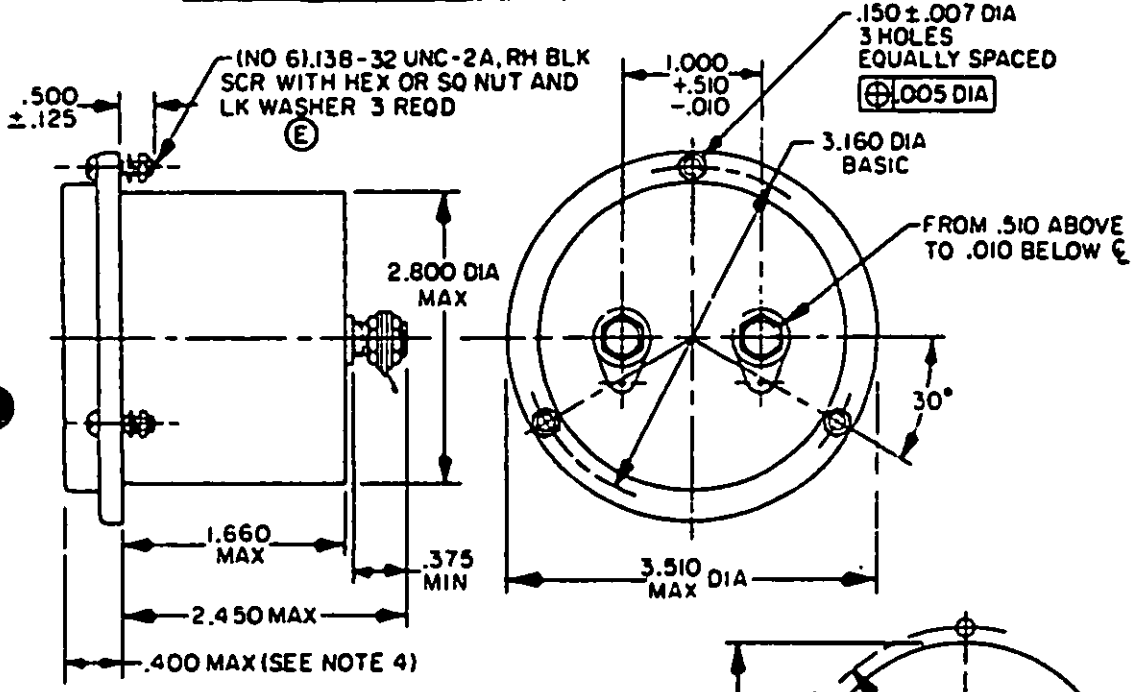
MILITARY SPECIFICATION SHEET

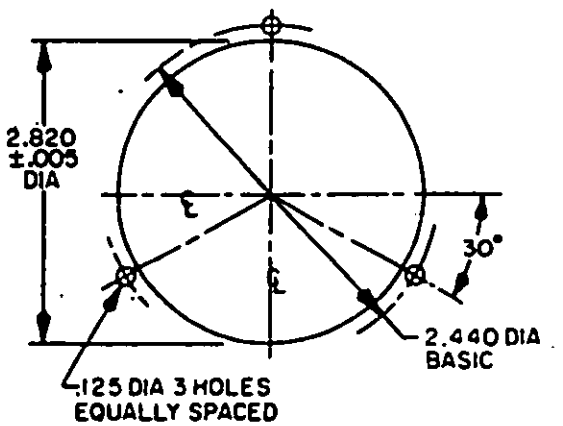
# METERS, ELECTRICAL INDICATING, PANEL TYPE, RUGGEDIZED: AMMETER, DC (FLUSH MOUNTING, ROUND FLANGE, 3-1/2 INCH) STYLE 38

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the meters described herein shall consist of this document and the latest issue of Specification MIL-M-10304.




| INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|
| .005 | .13 | .375 | 9.53 | 2.450 | 62.23 |
| .007 | .18 | .400 | 10.16 | 2.800 | 71.12 |
| .010 | .25 | .500 | 12.70 | 2.820 | 71.63 |
| .125 | 3.18 | .510 | 12.95 | 3.160 | 80.26 |
| .138 | 3.51 | 1.000 | 25.40 | 3.510 | 89.15 |
| .150 | 3.81 | 1.660 | 42.16 | | |




PANEL DRILLING PLAN

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Tolerance is ±1/2° on angles.
4. Maximum projection from front of panel including mounting gaskets (if used).

Ⓔ denotes changes

FSC 6625



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REQUIREMENTS:
(Readings expressed as a percent of full-scale value.)

Dimensions and configuration: See figure.

Full-scale ranges: See table I.

Scale:
Length: 2 inches minimum.
Pointer deflection: 90° minimum.

Position influence: ±2 percent change for 60° rotation from normal operating position.

Accuracy: ±2 percent.

(E) Overshoot: 40 percent maximum.

Response time: 2 seconds maximum.

Power consumption (loss) (at end scale deflection):
Microammeters and millammeters: 150 mV maximum.
Ammeters (self-contained): 75 mV maximum.
Ammeters (external shunt): 50 mV ±2 percent with leads of 0.065 ±0.010 ohms resistance.

High temperature cycling:
±4 percent.
±2 percent permanent change.

Temperature influence: ±1 percent change

Exposure to extreme temperatures:
±3 percent permanent change.

Overload capacity:
Momentary overload: ±2 percent permanent change.
Sustained overload:
±1 percent temporary zero shift.
±1 percent permanent zero shift.
±2 percent permanent change.

Dielectric withstanding voltage: 3,000 volts rms.

Vibration, random drop, and shock: Repeatability is not applicable for those microammeters below 100-microampere ranges.



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TABLE I. Full-scale ranges.

| Microamperes | | Milliamperes | | Amperes | | Kiloamperes 1/ |
|---|---|---|---|---|---|---|
| Zero left | Zero center | Zero left | Zero center | Zero left | Zero center | |
| 20 | 30-0-30 | 1 | 1-0-1 | 1 | 5-0-5 | 1.0 |
| 50 | 50-0-50 | 5 | 5-0-5 | 2 | 10-0-10 | 1.2 |
| 100 | 100-0-100 | 10 | 10-0-10 | 5 | 1/ 10-0-30 | |
| 200 | 200-0-200 | 30 | 30-0-30 | 10 | 1/ 20-0-20 | |
| 500 | 300-0-300 | 50 | 50-0-50 | 20 | 1/ 30-0-30 | |
| | 500-0-500 | 100 | 100-0-100 | 30 | 1/ 50-0-50 | |
| | | 200 | 200-0-200 | 1/ 50 | | |
| | | 300 | 300-0-300 | 1/ 100 | | |
| | | 500 | 500-0-500 | 1/ 200 | | |
| | | | | 1/ 300 | | |
| | | | | 1/ 500 | | |

1/ Used with external shunt.

Custodians:
Army - EL
Navy - EC
Air Force - 82

Review activities:
Army - EL, AV, AT, ME, WC
Navy - EC, MC
Air Force - 11
DSA - ES

User activities:
Army - MI
Navy - CG
Air Force - None

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 6625-0574-9)



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-M-10304/12E METERS, ELECTRICAL INDICATING, PANEL TYPE, RUGGEDIZED: AMMETER, DC (FLUSH MOUNTING, ROUND FLANGE, 3½ INCH) STYLE 36

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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